

Chapter 15 – Recovery Mode, Disk Structure & Troubleshooting

1. What is Recovery Mode in Windows?

Recovery Mode is a **special troubleshooting environment** in Windows that helps you fix system issues when your OS is not working properly.

When is it used?

- OS corruption 
 - Boot failure 
 - System not starting 
 - After bad update or driver issue 
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2. How to Enter Recovery Mode?

Method 1 (Automatic)

- When Windows fails to boot multiple times → it automatically opens Recovery Mode.

Method 2 (Manual)

- Turn ON laptop → when Windows logo appears → press power button forcefully 2–3 times 
- OR
- Go to:

Settings → System → Recovery → Advanced Startup → Restart Now

3. Recovery Mode Options Explained

After entering, you will see:

Continue

 Exit recovery and go back to Windows

Use a Device

 Boot from USB/DVD (used for OS repair or reinstall)

Troubleshoot (Most Important)

Contains:

- Reset This PC
- Advanced Options

Turn off your PC

 Shutdown system

4. Advanced Options (Core Troubleshooting Tools)

Startup Repair

- Fixes login/boot issues
- Works only for **minor problems (~50%)**

System Restore

 Use when:

- Installed app caused issue (e.g., antivirus crash scenario)

 Example:

- Installed antivirus → system corrupted
 - Restore to previous working state
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✖ Uninstall Updates

💡 Use when:

- Windows update caused slow system or crash

👉 Removes latest updates safely

⚙ Startup Settings

💡 Use when:

- Faulty app/service slowing system

👉 Disable startup programs from here

💻 Command Prompt (Powerful Tool) 💣

Used for **advanced repair using commands**

🔧 5. Important Repair Commands (Must Know for IT Support) ⚡

🔍 1. SFC Scan

sfc /scannow

📌 Purpose:

- Scans Windows system files
- Repairs corrupted files

🧠 Use Case:

👉 OS behaving weird, files missing

2. DISM Commands

✓ Check Health

DISM /Online /Cleanup-Image /CheckHealth

✓ Scan Health

DISM /Online /Cleanup-Image /ScanHealth

✓ Restore Health

DISM /Online /Cleanup-Image /RestoreHealth

Purpose:

- Fix deeper OS image issues (when SFC fails)

3. Check Disk (Hard Disk Repair)

chkdsk /f /r

chkdsk /f /r /x

Purpose:

- Fix disk errors
- Repair bad sectors

Disk Structure Concept

Component Meaning

Cluster Smallest storage unit 

Sector Group of clusters

Track Circular path on disk

Disk Full storage

 If cluster index breaks → file becomes **corrupt**

4. Boot Repair Commands

For MBR systems:

`bootrec /fixmbr`

For GPT systems:

`bootrec /fixboot`

Rebuild Boot Configuration:

`bootrec /rebuildbcd`

5. Disk Cleanup

`cleanmgr`

-  Clears temp/cache files
 -  Fixes login or indexing issues
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6. Network Troubleshooting Commands (Very Important)

Reset Network (Winsock)

`netsh winsock reset`

-  Resets network socket settings
-

Reset IP Configuration

`netsh int ip reset all`

-  Fixes IP-related issues
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Flush DNS Cache

ipconfig /flushdns

 Removes old DNS cache

Register DNS

ipconfig /registerdns

 Registers new DNS entries

Trace Route

tracert google.com

 Tracks path of packet (used for debugging network delays)

7. DNS Optimization (Speed Trick)

◆ Google DNS:

8.8.8.8

◆ Cloudflare DNS:

1.1.1.1

 Faster browsing 

8. Real-Life Troubleshooting Scenario

Problem: Internet Slow on One System

Steps:

1. Clear browser cache

Ctrl + Shift + Delete

2. Update network driver

3. Update BIOS
4. Change DNS to Google/Cloudflare
5. Run commands:
 - netsh winsock reset
 - netsh int ip reset
 - ipconfig /flushdns
6. Restart system 

9. Cache vs Cookies

Type	Meaning
Cache	Temporary files (images, scripts)
Cookies	Session/login data

Final Summary

- Recovery Mode = **System Rescue Tool** 
 - Commands = **Your weapons** 
 - Disk structure = **Foundation knowledge** 
 - Network commands = **Fast troubleshooting** 
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Q & A

1. BASIC / THEORY QUESTIONS

? Q1. What is Recovery Mode in Windows? 

✓ Answer:

Recovery Mode is a special troubleshooting environment in Windows used when the OS is not working properly.

👉 It helps to:

- Fix boot issues 
- Repair corrupted files 
- Restore system to previous state 
- Reset or reinstall OS 

💡 Simple Explanation:

It is like a **rescue mode for Windows** 

? Q2. When do you use Recovery Mode? 

✓ Answer:

We use Recovery Mode in situations like:

- System not booting 
- Blue Screen error (BSOD) 
- After bad update 
- Driver failure 
- OS corruption 

👉 Basically, when normal Windows cannot start.

? Q3. How do you enter Recovery Mode? 🔑

✓ Answer:

There are 3 methods:

✓ Automatic:

- System fails to boot multiple times → opens automatically

✓ Manual:

- Force shutdown during boot 2–3 times 🔁

✓ Settings:

- Settings → System → Recovery → Advanced Startup

? Q4. What are main options in Recovery Mode? 🚑

✓ Answer:

- Continue → Boot normally
- Use a Device → Boot from USB/DVD
- Troubleshoot → Main repair section 🔥
- Turn off PC

👉 Most important is **Troubleshoot**

2. ADVANCED OPTIONS QUESTIONS

? Q5. What is Startup Repair? ⚙️

✓ Answer:

Startup Repair automatically fixes:

- Boot issues
- Missing system files
- Startup errors

⚠️ Works only for **minor issues** (~50%)

? Q6. What is System Restore? 🔄

✓ Answer:

System Restore takes system back to previous working state.

👉 It restores:

- System files
- Installed apps

✗ Does NOT delete personal data

💡 Example:

Installed antivirus → system crashed → restore to earlier point

? Q7. What is “Uninstall Updates”? ✗

✓ Answer:

Used when Windows update causes:

- System crash
- Slow performance
- Boot issue

👉 It removes recent updates safely

? Q8. What is Startup Settings? ⚡

✓ Answer:

Used to:

- Disable startup apps
- Boot into Safe Mode
- Troubleshoot driver/service issues

? Q9. Why is Command Prompt important in Recovery Mode? 💣

✓ Answer:

Command Prompt allows advanced troubleshooting using commands.

👉 It is used for:

- Repair OS
- Fix boot issues
- Repair disk
- Reset network

💡 It is the **most powerful troubleshooting tool**

⚙️ 3. COMMAND-BASED QUESTIONS (VERY IMPORTANT 🔥)

? Q10. What is SFC command? 🔍

sfc /scannow

✓ Answer:

- Scans system files
- Repairs corrupted files

👉 Use when:

- System behaving abnormally
 - Missing/corrupt files
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? **Q11. What is DISM?** 🩺

✅ **Answer:**

DISM repairs deeper OS image issues.

Commands:

DISM /Online /Cleanup-Image /CheckHealth

DISM /Online /Cleanup-Image /ScanHealth

DISM /Online /Cleanup-Image /RestoreHealth

👉 Use when SFC cannot fix problem

? **Q12. What is CHKDSK?** 🗑️

chkdsk /f /r

✅ **Answer:**

- Fixes disk errors
- Repairs bad sectors

👉 Used when:

- Disk slow
 - File corruption
 - Disk errors
-

? Q13. What are Bootrec commands? 🧠

✓ Answer:

Used to fix boot issues:

bootrec /fixmbr

bootrec /fixboot

bootrec /rebuildbcd

👉 Fix:

- Bootloader issues
- OS not loading

? Q14. What is cleanmgr? 🧹

cleanmgr

✓ Answer:

Disk Cleanup tool:

- Deletes temp files
- Clears cache
- Improves performance

🌐 4. NETWORK COMMAND QUESTIONS

? Q15. What is netsh winsock reset? 🌐

✓ Answer:

Resets network socket settings.

👉 Fixes:

- Internet issues
- Network corruption

? Q16. What is ipconfig /flushdns? 🖌️

✅ Answer:

Clears DNS cache.

👉 Fixes:

- Website loading issues
- DNS errors

? Q17. What is tracert? 🗺️

tracert google.com

✅ Answer:

Tracks path of packet from system to server.

👉 Used for:

- Network delay analysis
- Routing issues

? Q18. What is DNS? 🌐

✅ Answer:

DNS converts:

👉 Domain → IP address

💡 Example:

google.com → IP address

? Q19. Which DNS servers improve speed? ⚡

✓ Answer:

- Google DNS → 8.8.8.8
- Cloudflare → 1.1.1.1

👉 Faster browsing 🚀

📁 5. DISK STRUCTURE QUESTIONS

? Q20. What is disk structure? 🧠

✓ Answer:

Disk is divided into:

- Disk → Full storage
- Track → Circular path
- Sector → Group of clusters
- Cluster → Smallest unit

👉 If cluster damaged → file corruption

? Q21. What is bad sector? ⚠️

✓ Answer:

A damaged area on disk where data cannot be stored.

👉 Fixed using:

- chkdsk /r
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6. REAL INTERVIEW SCENARIO QUESTIONS

 **Q22. System not booting — what will you do?** 

 **Answer:**

Step-by-step approach:

1. Enter Recovery Mode
2. Run Startup Repair
3. Try System Restore
4. Run:
 - sfc /scannow
 - DISM
5. Use bootrec commands
6. If not fixed → Reset PC

 Always follow structured approach

 **Q23. System slow after update?** 

 **Answer:**

1. Go to Recovery Mode
 2. Use “Uninstall Updates”
 3. Check drivers
 4. Clean temp files
 5. Restart system
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? Q24. Internet slow on one system? 🌐

✓ Answer:

Steps:

1. Clear browser cache
 2. Update network driver
 3. Change DNS
 4. Run commands:
 - netsh winsock reset
 - ipconfig /flushdns
 5. Restart
-

? Q25. Files getting corrupted frequently? 📁

✓ Answer:

1. Run chkdsk
 2. Check disk health
 3. Run SFC scan
 4. Replace disk if hardware issue
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? Q26. OS behaving weird (apps crashing)? 💥

✓ Answer:

1. Run SFC
 2. Run DISM
 3. Check updates
 4. Restore system
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? Q27. Laptop showing blue screen? ●

✓ Answer:

1. Enter Recovery Mode
2. Check Event Viewer (after boot)
3. Uninstall recent driver/update
4. Run SFC + DISM
5. Check hardware (RAM/Disk)

🔥 7. HR + PRACTICAL THINKING QUESTIONS

? Q28. How do you troubleshoot any issue? 🧠

✓ Answer:

My approach:

1. Identify problem 🔍
2. Ask user questions
3. Check logs/settings
4. Apply solution
5. Test system

👉 Focus on **root cause**

? Q29. Why do we use structured troubleshooting? 📋

✓ Answer:

Because:

- Saves time ⌚
- Avoids random fixes ❌

- Gives accurate solution 
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 **Q30. What tools do you use for troubleshooting?** 

 **Answer:**

- Task Manager
 - Event Viewer
 - Command Prompt
 - Device Manager
 - Recovery Mode
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